

Shehan Irteza Pranto

PHD CANDIDATE, DEPT. OF BIOMEDICAL INFORMATICS AND DATA SCIENCE, UNIVERSITY OF ALABAMA AT BIRMINGHAM

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RESEARCH INTERESTS

AI in healthcare; Deep learning; Agentic AI; Wearable Sensors; Biomedical Informatics

EDUCATION

PH.D. CANDIDATE, BIOMEDICAL AND HEALTH INFORMATICS

UNIVERSITY OF ALABAMA AT BIRMINGHAM

Birmingham, AL

Fall 2024 – [May 2028] (expected)

- **Thesis:** Continuous, Interpretable, and Personalized Deep Learning for Mortality Prediction in Critically Ill AKI Patients on CRRT
Advisors: Jin Chen, PhD; Javier A. Neyra, MD, MSCS. **Candidacy:** June 2026

M.S. IN ELECTRICAL ENGINEERING

THE UNIVERSITY OF ALABAMA

Tuscaloosa, AL

Aug 2022 – May 2024

- **Project:** Sensor-Based Monitoring of Vaping Behavior Using a Wearable Embedded System. **Advisor:** Edward Sazonov, PhD

B.SC. IN ELECTRICAL AND ELECTRONIC ENGINEERING

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Dhaka, Bangladesh

Apr 2016 – Jan 2021

- **Thesis:** Study on the Impact of VLC and FSO Technologies for the Development of 6G Communication Systems. **Advisor:** A.K.M. Ehtesanul Islam, PhD

RESEARCH EXPERIENCE

GRADUATE RESEARCH FELLOW

BIOMEDICAL INFORMATICS AND DATA SCIENCE, UNIVERSITY OF ALABAMA AT BIRMINGHAM

Birmingham, AL

Aug 2024 – present

- Designing deep learning models (Transformer, Multi-Instance Learning, Reinforcement Learning) on CRRT machine waveforms and EHR data to predict patient outcome in ICU patients with AKI.
- Developing a multi-agent LLM system that turns CRRT mortality model outputs into clinician-readable explanations, and a governance harness that flags unreliable predictions for clinician review.
- Collaborating with clinicians to provide insights for patient monitoring and treatment optimization.

GRADUATE RESEARCH ASSISTANT

CLAWS LAB (PROF. EDWARD SAZONOV), THE UNIVERSITY OF ALABAMA

Tuscaloosa, AL

Aug 2022 – May 2024

- **Hardware.** Programmed MSP430 firmware for a wearable sensor that records vaping puffs, assembled PCBs, and built device prototypes.
- **Software.** Developed the FRIENDS GUI and puff-detection software that turns wearable vaping-monitor logs into puff count, duration, and daily-use profiles for tobacco-use researchers (published, open source)

RESEARCH ENGINEER

AIMS LAB (PROF. KHONDAKER A. MAMUN), UNITED INTERNATIONAL UNIVERSITY

Dhaka, Bangladesh

Nov 2020 – Jun 2022

- **AI-Reception.** Built face recognition (ArcFace), speaker recognition (MFCC, GMM), Bangla ASR (DeepSpeech-2), and BERT dialogue modules for a hospital receptionist robot
- **PVDdoctor.** Developed voice and tremor signal processing and ML models to screen Parkinson's disease on a smartphone and estimate MDS-UPDRS severity
- **AIMScribe.** designed the speech-to-text and note-structuring pipeline for doctor-patient conversations
- **BRIGHT.** Co-wrote and budgeted a UK NIHR proposal on brain-robot stroke telerehabilitation with Ulster University and University of Essex

FUNDED PROJECTS

NIH/NIDDK Artificial Intelligence to Predict Outcomes in Patients with Acute Kidney Injury on Continuous Renal Replacement Therapy (R01DK133539). PhD researcher, UAB, 2024 – present

NIH/NIDDK The Relation of Soluble Klotho with Cardiovascular Disease, Chronic Kidney Disease Progression, and Blood Pressure in the Systolic Blood Pressure Intervention Trial (R01DK128208), PhD researcher, UAB, 2026

NIH/NCI FRIENDS: Automatic Detection of Vaping Rate and Patterning in the Lab and the Natural Environment Using Flexible, Robust Instrumentation of Electronic Nicotine Delivery Systems (R01CA266170). MS researcher, The University of Alabama, 2022 – 2024

UIU IAR AIMScribe: Multi-Lingual Ambient Clinical Intelligence with Localized Prescription Generation (UIU-IAR-01-2022-SE-24). Co-investigator, 2021 – 2024

ICT Division	PVDoctor: AI-Powered Cloud Platform for Parkinson’s Disease Screening and Monitoring. Research engineer, AIMS Lab, UIU, 2021
ICT Division	SMART Reception: An Artificial Intelligence Driven Bangla Language Based Receptionist System Employing Speech, Speaker, and Face Recognition for Automating Reception Services. Lead developer, AIMS Lab, UIU, 2021 – 2022

TEACHING & MENTORING EXPERIENCE

Fall 2026	One-Person Research Lab 101: Agentic AI for Research Instructor, Dept. of Biomedical Informatics and Data Science, UAB 5-week hands-on course on LLM agent systems: agent loops, multi-role teams with orchestrator and critic, PDF extraction with provenance, auditable research pipelines
Summer 2026	KUH KURE Undergraduate Research Experience Program, UAB Research Mentor Guided a research project on a safety layer that flags unreliable AI mortality predictions in AKI patients on CRRT for clinician review instead of automatic use

ACADEMIC HONORS

2025 – 2026	Blazer Graduate Research Fellowship, UAB Graduate School
2026	Abstract selected for oral presentation, ASN Kidney Week, session “Artificial Intelligence in Kidney Care”
2021	Top 10 of 1,051 teams, Mujib100 Idea Contest (1M BDT): Smartphone-based Parkinson’s disease screening
2021	Top 10 of 1,051 teams, Mujib100 Idea Contest (1M BDT): AI Customer Service, smart call center support

PROFESSIONAL AFFILIATIONS AND REVIEW ACTIVITIES

- Student Member, American Society of Nephrology (ASN)
- Reviewer, IEEE International Conference on Biomedical and Health Informatics (BHI) 2026: 5 abstracts
- Reviewer, IEEE International Conference on Bioinformatics and Biomedicine (BIBM) 2026, main track: 2 full papers

LEADERSHIP AND SERVICE

ASSISTANT STUDENT WELFARE SECRETARY BANGLADESH STUDENT ASSOCIATION, THE UNIVERSITY OF ALABAMA	Tuscaloosa, AL Feb 2023 – Jan 2024
• Coordinated housing and transportation support for incoming students and mentored them on academic and research planning	
RESEARCH VOLUNTEER RESEARCH REALM	Remote Mar 2020 – May 2020
• Survey design and data analysis on opportunities and challenges of telemedicine during COVID-19	

INDUSTRY EXPERIENCE

MACHINE LEARNING ENGINEER INTERN PIONEER ALPHA LIMITED	Remote Jul 2020 – Jan 2021
• Built production AI chatbots for clients end to end	

PUBLICATIONS

Total: 5 journal papers (3 published or accepted, 2 under review), 5 conference papers (4 published, 1 under review), 1 book chapter. Selected below; full list on Google Scholar.

Journal papers

1. L. J. Liu*, **S. I. Pranto***, J. Chen, and J. A. Neyra, “Artificial Intelligence in Acute Kidney Injury and Continuous Renal Replacement Therapy,” *Advances in Kidney Disease and Health*, accepted, 2026. *Equal contribution.
2. J. Yang, J. Kauffman, **S. I. Pranto**, T. Takeuchi, S. Goldstein, J. Chen, J. Lambert, G. N. Nadkarni, L. Chan, and J. A. Neyra, “Development and External Validation of a Machine Learning Model to Predict 7-Day CRRT-Free Survival in Patients on Continuous Renal Replacement Therapy,” *Intensive Care Medicine*, under review, 2026.
3. **S. I. Pranto**, L. W. Hawk, and E. Sazonov, “FRIENDS GUI: Data Collection and Visualization of Vaping Behavior from a Passive Vaping Monitor,” *Journal of Open Research Software*, vol. 14, 2026.
4. K. A. Mamun, R. A. Nabid, **S. I. Pranto**, S. M. Lamim, M. M. Rahman, N. Mahammed, et al., “Smart Reception: An Artificial Intelligence Driven Bangla Language Based Receptionist System Employing Speech, Speaker, and Face Recognition for Automating Reception Services,” *Engineering Applications of Artificial Intelligence*, vol. 136, 108923, 2024.

Conference papers

1. **S. I. Pranto**, J. Chen, and J. A. Neyra, “Rolling Day-Wise Mortality Prediction in Critically Ill Patients With AKI on CRRT Utilizing Machine Pressure Waveforms,” *IEEE International Conference on Big Data*, under review, 2026.
2. M. T. Ehsan, **S. I. Pranto**, and K. A. Mamun, “Real-Time Screening of Parkinson’s Disease Based on Speech Analysis Using Smartphone,” *10th International IEEE/EMBS Conference on Neural Engineering (NER)*, Italy, pp. 573–576, 2021.
3. **S. I. Pranto** et al., “Human-Robot Interaction in Bengali Language for Healthcare Automation Integrated with Speaker Recognition and Artificial Conversational Entity,” *3rd International Conference on Electrical & Electronic Engineering (ICEEE)*, Rajshahi, Bangladesh, pp. 13–16, 2021.

Book chapters

1. **S. I. Pranto**, “Security and Privacy for AR/VR Spaces,” in *Metaverse Platform Implementation*, CRC Press, pp. 164–182, 2025.

PRESENTATIONS

ASN Kidney Week	2026 , Denver, CO Oral: CRRT Machine Pressure Waveforms Improve Day-Wise ICU Mortality Prediction in Patients With AKI (session “Artificial Intelligence in Kidney Care”) 2025 , Houston, TX Poster, co-author: AKI Etiology and In-Hospital Mortality by Discharge Diagnosis Category
ATTIS Annual Translational and Transformative Informatics Symposium, UAB	2026 , Birmingham, AL Flash talk: Do CRRT Machine Signals Add Prognostic Value? A Multimodal Machine Learning Approach to ICU Mortality Prediction Poster: same title [2nd Place Poster Award] 2025 , Birmingham, AL Flash talk: Enhancing Prognostic Accuracy in AKI: A Systematic Data Quality Assurance Framework for CRRT Machine Data Poster: same title

TECHNICAL SKILLS

Programming	Python, R, Embedded C, MATLAB
Machine learning	PyTorch, TensorFlow, scikit-learn, RAG, Knowledge Graph
Microcontrollers	MSP430, STM32, PCB assembly
Design tools	Fusion 360 (3D printing), PSpice, Proteus, Cadence, OptiSystem
Visualization	Tableau, Power BI
Tools	Git, Linux, Windows, HPC (Slurm), LaTeX

REFERENCES

Jin Chen, PhD, FAMIA

Professor, Dept. of Medicine, Division of Nephrology; Vice Chair for Education, Dept. of Biomedical Informatics and Data Science, University of Alabama at Birmingham
jichen@uabmc.edu

Edward Sazonov, PhD

Professor, Dept. of Electrical and Computer Engineering; Director, CLAWS Lab, The University of Alabama
esazonov@ua.edu

Javier A. Neyra, MD, MSCS

Professor, Dept. of Medicine, Division of Nephrology, University of Alabama at Birmingham
jneyra@uabmc.edu

Khondaker A. Mamun, PhD

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